

The Macroeconomic Factors' Influence on Gold Futures Prices: A Case Study of U.S. Gold Futures

Table of Contents

1. Introduction	4
1.1. Background	4
1.2. Problem Statement	6
1.3. Research Questions	7
1.4. Objectives	7
2. Literature Review	8
2.1. Theoretical Review	8
2.1.1. Efficient Market Hypothesis	8
2.1.2. Liquidity Preference Theory	9
2.2. Empirical Review	10
2.2.1. The US Dollar Index	10
2.2.2. Consumer Price Index	10
2.2.3. Money Supply	11
2.2.4. Interest Rate	11
2.2.5. Industrial Production Index	12
3. Data and Methodology	13
3.1. Research Framework	13
3.2. Hypothesis Development	13
3.3. Data	14
3.4. Methodologies	15
3.4.1. Ordinary Least Squares	15
3.4.2. Multicollinearity Diagnostics	16
3.4.3. Heteroscedasticity Diagnostics	16
3.4.4. Autocorrelation Diagnostics	17
4. Empirical Results and Analysis	18
4.1. Descriptive Statistics	18

4.2.	Regression Analysis.....	19
4.3.	Diagnostics.....	20
4.3.1.	Multicollinearity Test.....	20
4.3.2.	Heteroscedasticity Test.....	21
4.3.3.	Autocorrelation Test	22
4.4.	HAC Standard Errors Model.....	24
4.5.	Lagged Dependent Variables Model.....	25
4.6.	Results Interpretation.....	27
5.	Conclusion.....	28
6.	Reference.....	30
7.	Appendix	39

1. Introduction

1.1. Background

Gold has long been regarded as a store of value and a safe haven during times of uncertainty. Throughout history, people have turned to gold when they're worried about inflation, currency depreciation, or economic instability (Baur & Lucey, 2010). While gold is widely used in jewellery and industrial applications, it also plays a significant role in financial markets. One way investors gain exposure to gold is through gold futures. A gold future is a standardised contract that allows someone to agree in advance to buy or sell a specific amount of gold at a set price on a certain date. These contracts are traded on exchanges like the CME Group, allowing investors to benefit from gold price movements without physically holding the metal (CME Group, n.d.).

Gold futures play a central role in the global financial system, serving multiple purposes. Speculators may use gold futures to attempt to profit from price changes, often without intending to take delivery of the gold. Instead, they buy and sell the contracts, hoping to close their positions before settlement (BullionVault, n.d.). On the other hand, hedgers use gold futures to hedge against risks like inflation or currency fluctuations, as the gold spot price (blue line) and the gold futures price (red line) are highly correlated as shown in Figure 1.1.1 (Baklaci et al., 2018; Jin et al., 2018). Gold futures are highly liquid and accessible. Investors can take positions without incurring storage or insurance costs, which would be necessary with physical gold. Additionally, gold has shown a low correlation with stocks and bonds, making it a valuable tool for portfolio diversification (Ghosh et al., 2004). Gold futures also support price discovery by consolidating real-time market data from a wide range of participants, including miners, jewellers, central banks, and institutional investors.



Figure 1.1.1: Gold spot (blue) vs futures (red) price. Source: Investing.com (n.d.a).

In recent years, gold prices have experienced a strong rally and reached their all-time high (Figure 1.1.2). Many analysts expect gold to continue breaking its highest record in the future (MCB Group, 2025). A mix of inflation fears, geopolitical tensions, and rising demand from central banks is driving this increase. Countries such as China and India are increasing their gold purchases as they seek to diversify their currency holdings and reduce their reliance on the US dollar. Meanwhile, prolonged periods of loose monetary policy have raised concerns about currency stability. In this environment, gold futures remain an essential tool for investors to manage risk and navigate today's complex financial landscape. Thus, studying the macroeconomic variables' impact on gold futures price is essential.



Figure 1.1.2: Gold prices historical data from 2015 to 2025. Source: Investing.com (n.d.b)

1.2. Problem Statement

Gold futures prices have risen significantly in recent years. Notably, in April 2025, gold experienced its highest single-day gain since the pandemic-induced market turmoil of 2020, underscoring its sensitivity to global economic volatility (WSJ Staff, 2025). These sharp fluctuations highlight gold's dual significance as a safe-haven asset during crises and an inflation hedge, making its price behaviour crucial for investors and policymakers. However, the complicated relationship between macroeconomic dynamics and gold futures prices remains imperfectly understood, warranting further investigation.

A complex network of economic indicators drives the price of gold. The US dollar index has a well-documented inverse relationship with the price of gold. When the dollar falls, gold becomes cheaper for foreign investors, driving up demand and prices. Meanwhile, a stronger dollar suppresses the appeal and price of gold (Swissbullion, 2025). Similarly, inflation has a direct effect on gold's appeal, increasing inflation erodes purchasing power and decreases currency value, causing investors to consider gold as a store of wealth (Honor, 2024). Monetary policy also plays a key role. For instance, expansions in money supply depreciate currencies, prompting investors to seek gold as a hedge (Stetler, 2025). In contrast, interest rate hikes typically lower gold prices because the yield-bearing assets, such as bonds and savings are more attractive. Conversely, lower interest rates boost gold demand (StonexBullion, 2025). Lastly, the industrial production index has displayed a positive relationship with gold prices, due to gold's industrial usage or demand during economic overheating (GoldPriceForecast, n.d.). Despite these findings, the variables' relationship with the gold price remains unexplored in today's market environment, especially under this rapidly changing economic landscape. Therefore, further study with latest data is necessary.

Market players must understand these dynamics. Institutional investors, hedge funds, and traders rely on accurate gold price forecasts to optimise portfolios, manage risk, and identify arbitrage opportunities. Improved predictive models could be helpful tools for central banks to forecast the impact of policy changes on commodity markets. According to He (n.d.), gold futures prices are influenced by a complex set of macroeconomic and financial factors, making their volatility crucial for investment decisions and market stability. This study aims to overcome that gap by identifying the significant predictors and analysing how key macroeconomic factors influence US gold futures prices.

1.3. Research Questions

1. How does the US dollar index impact the gold futures prices?
2. How does the Consumer Price Index influence the gold futures price?
3. How does the money supply (M2) affect the gold futures price?
4. How do interest rates impact the gold futures price?
5. How does the industrial production index affect the gold futures price?

1.4. Objectives

1. To analyse the effect of the US dollar index on the gold futures price.
2. To investigate the influence of the Consumer Price Index on the gold futures price.
3. To examine the impact of money supply on gold futures price.
4. To evaluate the effect of interest rates on the gold futures price.
5. To determine the industrial production index's effect on the gold futures price.

2. Literature Review

2.1. Theoretical Review

2.1.1. Efficient Market Hypothesis

The Efficient Market Hypothesis (EMH) posits that market prices should reflect all available information. The EMH is separated into weak-form, semi-strong-form, and strong-form (Țițan, 2015). Under the condition of an efficient market, investors cannot create excess returns consistently (Fama, 1991). The only way investors can increase their returns is by increasing their risk exposures (Team, 2024).

The EMH can be applied to the gold market, which is historically a marketplace for risk hedging. This has made gold an attractive investment during economic crises, resulting in a surge in gold prices during economic downturns. According to Ntim et al. (2015), the gold market in developed countries has exhibited the weak form EMH, which means that the gold price has reflected all publicly available historical information (Lucey et al., 2012). However, private information is not reflected in the price, and the future gold price cannot be predicted using past information. In fact, the publicly available macroeconomic data should have been reflected in the gold price, and any new information should be incorporated into the price immediately when it becomes available (Wang et al., 2024).

Ntim et al. (2015) argued that the gold market does not fully conform to the Random Walk Hypothesis as macroeconomic variables influence the gold price. Chang (2024) provided evidence for this phenomenon by highlighting that price fluctuations in the gold market reflect a large amount of relevant information. Moreover, Raju and Shirodkar (2020) challenged the Random Walk Hypothesis by finding that the spot price and futures price have a causal relationship with each other. Hence, the gold futures price is influenced by and can be predicted using past and current information. The volatility of macroeconomic variables not only affects the spot market but also has an indirect impact on the future market (Ntim et al., 2015). Therefore, investors can predict and potentially benefit from the gold futures price by leveraging information from the spot market. Thus, our study will challenge the Random Walk Hypothesis by investigating whether the gold futures price is determined by past information, such as its previous period price.

2.1.2. Liquidity Preference Theory

Liquidity Preference Theory (LPT) is a macroeconomic theory that explains why people tend to hold highly liquid assets for trading, precautionary, and speculative motives (Ogiriki & Andabai, 2014). LPT explains that the asset investors prefer depends on different economic conditions (Bibow, 2013). The theory demonstrates a negative relationship between currency demand and interest rate. Market conditions, such as inflation, that trigger interest rate adjustments affect the demand for liquidity assets, and thus, the price of these assets.

Gold acts as a secured collateral and liquidity asset in the financial market, perceived as an interest-free asset that has a competitive relationship with interest-earning assets and cash (Anthina, 2024). In times when holding money is undesirable, such as during periods of high inflation, gold is often considered the best investment to preserve value (Abdullah & Bakar, 2015). This led to a surge in demand for gold during periods of inflation and rising gold prices. According to research, a negative relationship exists between the interest rates, US Dollar Index, and the gold price (Jia et al., 2023). If the Central Bank decides to increase the interest rate, it will cause the opportunity cost of holding gold to rise, leading to a decrease in the gold price. Higher interest rates will also appreciate the dollar, making gold, which is quoted in dollars, more expensive for foreign investors. This will decrease the demand for gold and lower its price.

When the market faces economic instability and currency depreciates, investors tend to hold more gold to hedge against currency risk (Oka, 2021). For example, the gold price has rocketed since investors preferred to hold gold during the 2008 Global Financial Crisis (Hergt, 2013). During times of economic crisis, the central bank adjusts the interest rate to lower the inflation rate or increases the money supply to foster economic growth. The interest rate adjustment results in a change in the currency's value. As currency appreciates, gold investments become less attractive, and the gold price decreases. Therefore, the Central Bank's monetary policy decisions affect gold prices by shifting the demand and supply of liquid money (Ergül & Karakaş, 2024). Due to the characteristics of gold, the behaviour of investors holding gold is driven by both precautionary motives and speculative motives as proposed in the LPT.

In summary, LPT explains the framework of investors' behaviour towards holding liquid money based on changes in economic variables, which in turn affects gold investment behaviour and the gold price (Shubik & Yao, 1990).

2.2. Empirical Review

2.2.1. The US Dollar Index

The US dollar index is a composite that tracks the relative strength of the US dollar against a basket of major currencies. It usually reflects the US real exchange rate. The USD exchange rate negatively impacts the gold price, as a stronger USD may make gold more expensive for foreign investors, thereby reducing its demand and leading to a lower gold price (Levin et al., 2006; Lampinen, 2007; Batten et al., 2010; O'Connor and Lucey, 2012; Choong et al., 2012; Fan et al., 2014; Sun, 2018; Chiang, 2022). Conversely, when the dollar depreciates due to economic uncertainty, investors may increase their investment in gold as a store of value, driving the gold price up (Jabeur et al., 2024). Mo et al. (2017) provide evidence to the argument by discovering that the USD real effective exchange rate has a significant negative impact on gold futures price in the Chinese market. Additionally, Wang et al. (2019) found a negative relationship between the US dollar index and the gold futures price. In contrast, the US Dollar Foreign Exchange Rate Index Volatility has been shown to have a significant positive impact on low-frequency volatility in China's and the U.S. gold futures market (Liu et al., 2015). Besides, Dinh et al. (2022) found a significant negative relationship between exchange rates and gold futures prices in both the US and global markets during the post-2008 global financial crisis period. Zhang and Qin (2022) argued that the RMB exchange rate has limited influence on China's gold price.

2.2.2. Consumer Price Index

The consumer price index (CPI) is a popular indicator to assess an economy's inflation. Levis et al. (2006) had proven a long-term one-for-one relationship between the gold price and the U.S. general price level. Another study supports the existence of both long-run and short-run relationships between gold prices and the US general price level (Lampinen, 2007). Inflation rates are found to have a negative influence on gold prices (Ibrahim et al., 2014; Hashim et al., 2014). Conversely, Choong et al. (2012) conclude that the world CPI has a positive influence on gold prices. Moreover, several studies have demonstrated that CPI has a significant positive impact on gold futures prices (Batten et al., 2010; Silva, 2014; Liu et al., 2015; Mo et al., 2017; Risse, 2019; Valadkhani et al., 2022; Radev et al., 2023). Additionally, inflation's positive impact on the long-run volatility of gold price during the post-global financial crisis period is highlighted in Fang et al.'s (2018) study. Tomaran et al. (2011), Dierinck et al. (2012), Kamran et al. (2014), Erdoğan (2017), and Ambarita and Djamaluddin (2020) conclude that there is no

relationship between the inflation rate and gold price. Rana's (2022) multi-regional study has discovered a positive relationship between CPI and gold price in Australia and India, a negative relationship in Japan, and an insignificant relationship in China.

2.2.3. Money Supply

Money supply (M2) is the volume of circulating money and readily spendable funds in the economy. M2 growth is often associated with expansionary monetary policy, which leads to devalued fiat currencies and increases demand for gold. A cointegrating relationship is identified in Tantatape and Komain's (2007) and Synek's (2024) study between money supply and gold prices. The study emphasised that increases in money supply lead to higher gold prices in the long run. Artigas (2010) and Kasteler (2017) discovered a positive relationship between M2 and gold price. According to Lucey et al. (2016), gold serves as a good hedge against inflation in the U.S. and the U.K. due to its positive correlation with the money supply. Additionally, Batten et al. (2010) suggest a positive relationship between money supply and gold futures price. In contrast, Seemuang and Romprsert (2013) and Sipkova and Sipko (2014) found a negative relationship between money supply and gold price. Consistently, research conducted in Japan (Lucey et al., 2016) and India (Mo et al., 2017) has revealed a negative relationship, where an increasing money supply has led to decreasing gold prices. Nylund (2017), however, concludes that the money supply does not affect the gold price.

2.2.4. Interest Rate

The interest rate in the U.S. economy is called the federal funds rate. It is a monetary policy tool used to control inflation. Researchers had suggested a negative relationship between the interest rate and gold price, as higher interest rates lead to a higher opportunity cost of holding non-yielding gold, thereby reducing its demand (Batten et al., 2010; Wang & Chueh, 2012; Zhang & Ci, 2020; Dinh et al., 2022; Ben et al., 2024). Additionally, Erb and Harvey (2013), Mamcarz (2015), and Hashim et al. (2017) suggest a negative relationship between real interest rates and the real gold price. Valadkhani et al. (2022) identified an inverse relationship between the ten-year Treasury interest rate and gold price. Rana (2022) examined the relationship between interest rates and the gold price, producing different conclusions in various countries. A short-run negative relationship is identified in Japan, whereas a positive relationship is observed in China. Some research support the statement that the interest rates have a positive and significant impact on gold futures prices (Mo et al., 2017; Wang et al., 2019; Rana & Connor, 2024). On the other hand, Lampinen (2007) concludes that the current physical interest

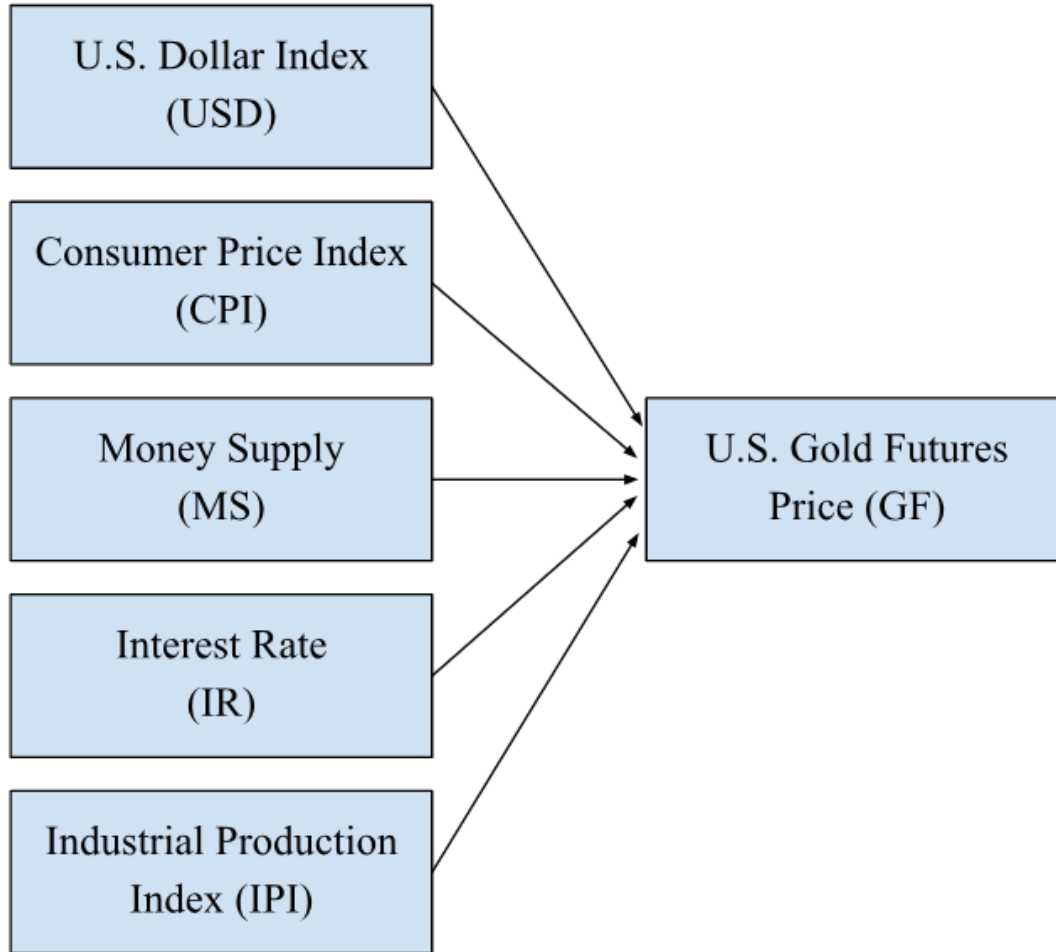
rate does not affect the gold prices. Some researchers support this conclusion by stating that the interest rates do not affect the gold price (Tomaran et al., 2011; Sindhu, 2013; Kamran et al., 2013; Sun, 2018; Ambarita & Djamaluddi, 2020).

2.2.5. Industrial Production Index

The industrial production index (IPI) measures the real output of manufacturing, mining, and utilities sectors within an economy over time, reflecting changes in production volume after adjusting for inflation. Tantatape and Komain's (2007) study found that IPI has a cointegrating relation with the gold price. Kim (2023) found that gold prices have been relatively flat, except during economic booms, while industrial production has a positive impact on financial indicators. This highlights the delicate dynamics between industrial output and the gold market. Mo et al. (2017) identified a positive relationship between IPI and the gold futures price. Similarly, Singh and Kaur (2020) and Gold Price Forecast (2019) found a positive correlation between IPI and gold prices, suggesting that increases in industrial production lead to a rise in gold prices due to inflationary pressures. Additionally, Rana (2022) found a positive relationship between industrial production and the gold price in Japan, but a negative relationship in India. The negative relationship is proven in Rana and O'Connor's (2023) study, which observed that in emerging markets, higher industrial production is associated with lower gold prices. They suggest that increased industrial activity may make gold less attractive as an investment in these economies, as other investments will be better rewarding during economic bloom. In contrast, Simakova (2011) argued that the relationship between IPI and gold price is statistically insignificant.

3. Data and Methodology

3.1. Research Framework



3.2. Hypothesis Development

H1: The U.S. Dollar Index significantly influences the U.S. Gold Futures Price.

H2: The Consumer Price Index significantly influences the U.S. Gold Futures Price.

H3: Money Supply significantly influences the U.S. Gold Futures Price.

H4: Interest Rate significantly influences the U.S. Gold Futures Price.

H5: Industrial Production Index significantly influences the U.S. Gold Futures Price.

Hypothesis Equation:

$$GF = \beta_0 + \beta_1 USD + \beta_2 CPI + \beta_3 MS + \beta_4 IR + \beta_5 IPI + \varepsilon$$

3.3. Data

The data used in this study is monthly data spanning 10 years from March 1, 2014, to March 1, 2024, with a total of 121 data points. The number of data points after adjusting for lagged effects will be 120.

The primary dependent variable is the Gold Futures Price (GF). The data is measured based on the settlement price reported from the most active exchange, the COMEX gold futures contract. The prices, reported in U.S. dollars per ounce, are derived from daily settlement prices, which in turn are based on the volume-weighted average price (VWAP) for the closing period on a given trading day. Monthly averages are used in this study to smooth out short-term volatility and better reflect macroeconomic trends.

The U.S. Dollar Index (USD) measures the strength of the U.S. dollar relative to a basket of 26 major foreign currencies, using January 2006 as the base year (Gurufocus.com, 2025). The nominal bilateral exchange rates are collected from Bloomberg. Each currency's weight in the index is determined using the geometric mean of the currencies' trade shares with the U.S., which are updated annually.

The Consumer Price Index (CPI) captures inflationary trends, specifically the growth rate of the U.S. all-items CPI for urban consumers' consumption of a basket of goods and services. The CPI is calculated by the U.S. Bureau of Labour Statistics using the Laspeyres index, which tracks the price of the same basket of products and services over time. The growth rate is simply the percentage change compared to the previous period. The data is published monthly and indicates the trend in general price levels (FRED, n.d.b.).

The M2 aggregate measures the Money Supply (MS) in billions of dollars. Before May 2020, the M2 aggregate consisted of M1 plus savings deposits and small-denomination time deposits, less individual retirement account and Keogh balances at depository institutions, and balances in retail money market funds (MMFs) less IRA and Keogh balances at MMFs (FRED, n.d.c). After May 2020, M2 equals M1 plus small-denomination time deposits less IRA and Keogh balances at depository institutions, and balances in retail MMFs less IRA and Keogh balances at MMFs.

The Interest Rate (IR) is captured through the Federal Funds Effective Rate, which is the average interest rate on overnight federal funds traded among the U.S. depository institutions (FRED, n.d.d). The rate is entirely determined by the borrower and lender of federal funds, and

the weighted average of the rate charged is the effective federal funds rate. The daily rate is calculated by the Federal Reserve using transaction data from a broad panel of banks, and the monthly average rate is used in this study.

Finally, the Industrial Production Index (IPI) serves as a proxy for real economic activity in the manufacturing, mining, and utilities industries. It measures the real output of all relevant U.S. establishments, regardless of ownership (FRED, n.d.e). The index is a seasonally adjusted Fisher Ideal Index calculation with 2017 as the base year. The weightage in the calculation is derived from each industry's share of GDP.

Variable	Measurement	Source
GF	Monthly closing price of gold futures	(Investing.com, 2025)
USD	Monthly average value of the U.S. Dollar Index	(FRED, n.d.a)
CPI	Growth Rate of Monthly U.S. CPI: All items: Total for U.S.	(FRED, n.d.b)
MS	M2 money supply	(FRED, n.d.c)
IR	Federal Funds Effective Rate	(FRED, n.d.d)
IPI	Monthly U.S. Industrial Production Index	(FRED, n.d.e)

Table 3.3: Summary of the variables.

3.4. Methodologies

3.4.1. Ordinary Least Squares

In this study, Ordinary Least Squares (OLS) is applied to estimate the relationship between the GF and the macroeconomic data. OLS is a fundamental econometric method used to obtain unbiased and efficient estimators based on classical linear regression assumptions. However,

several assumptions must be satisfied to ensure that the OLS estimators are reliable, including that the independent variables should not exhibit high multicollinearity, the variance of the error terms should remain constant across observations, and the error terms should be uncorrelated with one another over time. Therefore, to ensure the accuracy of the model, this study conducted a series of diagnostic tests to detect the presence of multicollinearity, heteroskedasticity, and autocorrelation, which will be discussed in the following sections.

3.4.2. Multicollinearity Diagnostics

If the independent variables are highly correlated with each other, a multicollinearity problem exists. When two or more variables move together too tightly, it becomes difficult to tell which one is truly influencing the dependent variable. This overlap can inflate the standard errors of the regression coefficients, making some variables appear statistically insignificant when they actually matter. Consequently, multicollinearity makes it difficult and unreliable to interpret the individual impact of each variable in the model. Two standard methods are employed to investigate this issue. First, a correlation matrix was created to examine the strength of the relationships between the independent variables. Second, a variance inflation factor (VIF) was used for testing. This method involves running a series of auxiliary regressions that regress a regressor on the remaining regressors. The R-squared value will be used to calculate the VIF, where a VIF value above 10 indicates that the variable is contributing to a multicollinearity problem.

3.4.3. Heteroscedasticity Diagnostics

Heteroscedasticity occurs when the variances of the disturbance terms in the classical linear regression models are non-standard, but the error varies with the values of the independent variables. In the present study, macroeconomic variables can vary over a research period, leading to inequalities in prediction errors in the price of gold futures and a heteroscedasticity issue. In the case of heteroscedasticity, although the estimates provided by OLS of the coefficients remain unbiased, the standard errors become inappropriate, which in turn influences the t-statistic and p-values, leading to incorrect conclusions. Hence, this study will detect the heteroscedasticity problem using the graphical method, Breusch-Pagan-Godfrey test, and White's test.

3.4.4. Autocorrelation Diagnostics

Autocorrelation refers to the correlation between the error in one period and the error term in a previous period, countering the assumption of error term independence. The gold futures price and macroeconomic variables with time trend features may have current price movements that depend on past values and result in systematic patterns of residuals. In an autocorrelated case, OLS estimators are unbiased; however, the standard errors are underestimated, and t-statistics are overestimated, resulting in biased outcomes in a test of significance and a reduction in the accuracy of the prediction. This study will employ the graphical method, the Durbin-Watson test, the correlogram of squared residuals, and the Breusch-Godfrey test to detect autocorrelation.

4. Empirical Results and Analysis

4.1. Descriptive Statistics

GF exhibits substantial volatility (Std Dev = 314.6340, Range = 1,060.20 - 2,227.10) and right-skewed distribution (Skewness = 0.390485), its platykurtic nature (kurtosis = 1.648291) indicates fewer extreme outliers than a normal distribution. Such platykurtic nature aligns with the gold's nature as a risk hedge investment. The significant Jarque-Bera statistic at p-value of 0.002 confirms non-normality, justifying the use of robust estimation methods. As for the independent variables, the data varies significantly as observed from data range and standard deviation. Furthermore, CPI, MS, and IR are right-skewed, while USD and IPI is left-skewed. The Kurtosis suggest that most independent variables is leptokurtic, except MS that is platykurtic. Moreover, except for CPI, all independent variables' Jarque-Bera statistics are significant, indicating non-normal distribution of the data.

	GF	USD	CPI	MS	IR	IPI
Mean	1515.094	113.4488	0.236810	16146.12	1.397603	100.7898
Median	1327.300	114.4106	0.215078	14526.20	0.650000	101.6718
Maximum	2227.100	127.4828	1.373608	21749.60	5.330000	104.1038
Minimum	1060.200	93.60390	-0.668694	11219.10	0.050000	84.68120
Std Dev	314.6340	7.046144	0.363256	3725.859	1.656724	3.018109
Skewness	0.390485	-1.058257	0.339990	0.326980	1.294159	-2.586150
Kurtosis	1.648291	4.356005	3.628129	1.457508	3.509228	12.93454
Jarque-Bera	12.28671	31.85516	4.320290	14.15169	35.08346	632.4655
Probability	0.002148	0.000000	0.115308	0.000845	0.000000	0.000000

Table 4.1: Descriptive Statistics.

4.2. Regression Analysis

The regression results are shown in Table 4.2. Four significant relationships have been identified in the regression analysis. Firstly, the results support H1 by suggesting a negative relationship between USD and GF with a p-value of less than 0.01. In contrast, CPI is found to be insignificant in the analysis; therefore, H2 is rejected. Additionally, a significant positive relationship exists between MS and GF, with a p-value of less than 0.01. Hence, H3 is supported. Additionally, H4 is supported as IR has a significant positive influence on GF, with a p-value of less than 0.01. Lastly, IPI has a significant negative relationship with GF at the level of significance of 0.01. Thus, H5 is supported. The model exhibits strong explanatory power, with an R-squared value of 0.924967 and an adjusted R-squared value of 0.921705. However, the model may be subject to problems with multicollinearity, heteroscedasticity, and autocorrelation. Thus, the diagnostics on the model are essential.

Variables	Coefficient	Std. Error	T-statistic	p-value
C	4486.428	469.0723	9.564470	0.0000***
USD	-13.98623	1.904190	-7.344974	0.0000***
CPI	10.25255	25.61427	0.400267	0.6897
MS	0.084295	0.003389	24.87482	0.0000***
IR	60.37884	7.581121	7.964369	0.0000***
IPI	-28.10266	3.495941	-8.038654	0.0000***
R-squared	0.924967	Mean dependent var		1515.094
Adjusted R-squared	0.921705	S.D. dependent var		314.6340
F-statistic	283.5334	Durbin-Watson stat		0.698711
Prob(F-statistic)	0.000000	AIC		11.84174

Notes: ***p-value < 0.01.

Table 4.2: Regression Analysis.

4.3. Diagnostics

4.3.1. Multicollinearity Test

Table 4.3.1.1 shows the pair-wise correlations among the variables. Since there is no high correlation between the independent variables, the model is less likely to encounter multicollinearity problems. However, further statistical investigation is necessary to make an accurate conclusion.

	GF	USD	CPI	MS	IR	IPI
GF	1.000000					
USD	0.579904	1.000000				
CPI	0.373673	0.105459	1.000000			
MS	0.931052	0.683209	0.410782	1.000000		
IR	0.473781	0.545068	0.010172	0.447966	1.000000	
IPI	-0.082224	-0.135748	0.068306	0.001665	0.447772	1.000000

Table 4.3.1.1: Correlation Matrix

Multiple auxiliary regressions are performed using each independent variable as the dependent variable and the remaining independent variables as the regressors. The R-squared values of the auxiliary regressions are extracted to calculate the variance inflation factor (VIF), which determines whether a multicollinearity problem exists. The outcomes indicate that none of the VIFs exceed 10, suggesting the absence of a multicollinearity problem.

Dependent Variable	Auxiliary Regression R²	VIF
USD	0.641210	2.787146
CPI	0.253940	1.340375
MS	0.594839	2.468155

IR	0.590554	2.442325
IPI	0.419816	1.723591

Table 4.3.1.2: Summary on the Variance Inflation Factor (VIF).

4.3.2. Heteroscedasticity Test

The graphical method, as shown in Figures 4.3.2.1 and 4.3.2.2, visualised the pattern of the squared residuals. Following the histogram, the frequency distribution of the squared residuals is skewed to the right, suggesting a possible heteroscedasticity problem. Additionally, the scatter plot of squared residuals against predicted regressand displays systematic patterns, suggesting a heteroscedasticity problem.

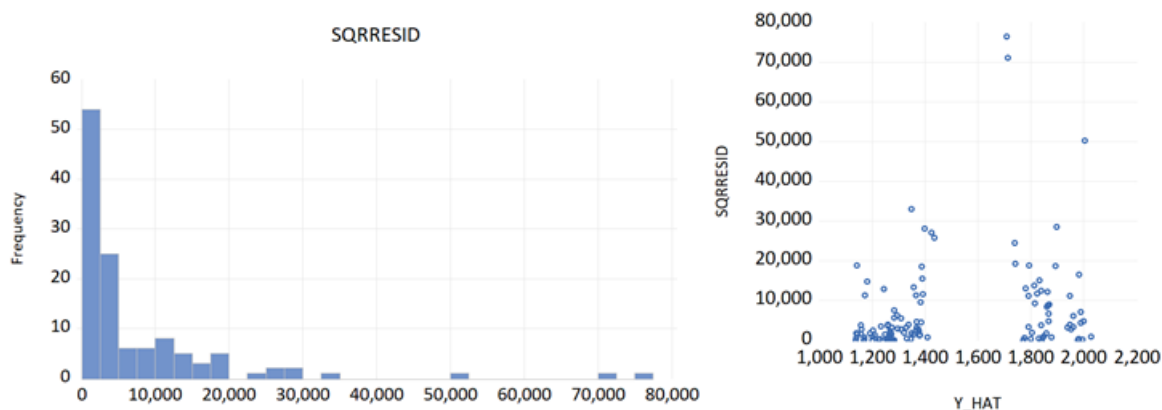


Figure 4.3.2.1 & 4.3.2.2: Histogram of distribution of the squared residuals and scatter plot of squared residuals against predicted dependent variable.

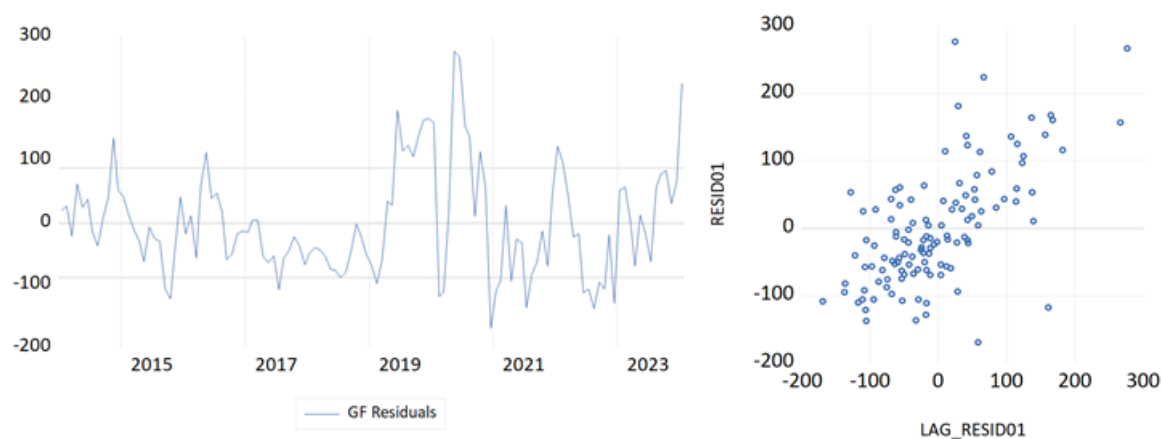
The summary of the two test methods employed to detect the heteroscedasticity problem is illustrated in Table 4.3.2.3. The Breusch-Pagan-Godfrey test generates a significant F-statistic, supporting the presence of a heteroscedasticity problem. Similarly, the White's test produced a significant F-statistic, conforming the conclusion in the Breusch-Pagan-Godfrey test. Thus, remedial actions must be taken to resolve the problem.

Test Methods	F-statistic	Prob(F)
Breusch-Pagan-Godfrey Test	3.410511	0.0066
White's Test	3.627257	0.0000

Table 4.3.2.3: Summary of the heteroscedasticity tests.

4.3.3. Autocorrelation Test

Graphs 4.3.3.1 and 4.3.3.2 display the visual representation of the residuals against time and lagged residuals, respectively. The residuals exhibit non-random patterns chronologically, and a positive relationship is observed between the residuals and the lagged residuals, as identified in the scatter plot. Therefore, the graphical method proposed an autocorrelation problem in the model.



Graph 4.3.3.1 & 4.3.3.2: Chronological line chart of residuals and scatter plot of residuals against lagged residuals.

Table 4.3.3.3 presents a summary of the Durbin-Watson test results for the model. The Durbin-Watson statistic is 0.698711. For our study, there are 121 observations and 5 independent variables. Hence, the dL is 1.610 and the dU is 1.794. As 0.698711 is lower than the dL, there is a strongly positive autocorrelation in the model. The correlogram of residuals squared in Graph 4.3.3.4 has identified autocorrelations in multiple lags of up to lag 7. The Q-statistics, along with the extremely low p-values, strongly reject the null hypothesis that there is no autocorrelation. Nevertheless, the exact level of autoregression requires further investigation. Thus, the Breusch-Godfrey Serial Correlation Lagrange Multiplier (LM) Test will be conducted to determine the autoregressive order for the model.

Durbin-Watson statistic (DW)	0.698711
n = 121	k = 5
dL = 1.610	dU = 1.794
Decision rule	Reject H_0 if $DW < 1.610$ or $DW > 2.390$
Decision	$0.698711 < 1.610$. Reject H_0 and conclude that there is a positive autocorrelation

Table 4.3.3.3: Durbin-Watson Test Summary.

Sample: 2014M01 2024M01
Included observations: 121

Autocorrelation	Partial Correlation		AC	PAC	Q-Stat	Prob
		1	0.454	0.454	25.514	0.000
		2	0.228	0.027	31.989	0.000
		3	0.120	0.009	33.801	0.000
		4	0.194	0.166	38.580	0.000
		5	0.271	0.159	48.001	0.000
		6	0.335	0.174	62.497	0.000
		7	0.317	0.118	75.640	0.000
		8	0.089	-0.169	76.677	0.000
		9	0.062	0.018	77.190	0.000
		10	0.023	-0.082	77.263	0.000
		11	0.192	0.139	82.265	0.000
		12	0.092	-0.151	83.415	0.000
		13	0.098	0.031	84.736	0.000
		14	0.021	-0.034	84.797	0.000
		15	-0.038	-0.043	85.005	0.000

Graph 4.3.3.4: Correlogram of residuals squared

The result summary on the Breusch-Godfrey Serial Correlation LM Test is displayed in Table 4.3.3.5. As suggested in the correlogram of residuals squared, a total of 8 lagged residuals are included in the model to determine the autoregressive order. The F-statistic is highly significant in the test, with a p-value of less than 0.001. Regarding the lagged residuals, only the first lagged residual is statistically significant, indicating that the model exhibits first-order autocorrelation. Thus, the remedial measure will be based on the AR(1) scheme.

F-statistic	283.5334	Prob.F	0.0000	
Obs*R-squared	56.42882	Prob. Chi-Square(8)	0.0000	
Lagged Residuals	Coefficient	Std. Error	T-statistic	p-value
Resid(-1)	0.726983	0.097079	7.488547	0.0000
Resid(-2)	-0.023166	0.120267	-0.192625	0.8476
Resid(-3)	-0.143017	0.118162	-1.210350	0.2288
Resid(-4)	0.193608	0.120280	1.609638	0.1104
Resid(-5)	0.092883	0.119610	0.776545	0.4391
Resid(-6)	-0.185300	0.118811	-1.559621	0.1218
Resid(-7)	0.228880	0.120647	1.897116	0.0605
Resid(-8)	-0.081043	0.106779	-0.758980	0.4495

Table 4.3.3.5: Breusch-Godfrey Serial Correlation LM Test summary.

4.4. HAC Standard Errors Model

As we have detected both heteroscedasticity and autocorrelation problems in the model, we should switch the model covariance method to HAC Standard Errors (Newey-West) to estimate the model more accurately. Under this method, the variables' standard errors, t-statistics, and p-values change, but the conclusion remains the same. Four variables —USD, MS, IR, and IPI— are statistically significant at a 1% level of confidence. However, their standard errors increased, causing T-statistics to decrease in absolute terms. This causes the significance to weaken, signalling that the original OLS results are overconfident. Nevertheless, the model still supports that USD, MS, IR, and IPI are significant macroeconomic determinants influencing the gold futures price.

Variables	Coefficient	Std. Error	T-statistic	p-value
C	4486.428	764.4916	5.868512	0.0000***
USD	-13.98623	2.727590	-5.127686	0.0000***
CPI	10.25255	27.76045	0.369322	0.7126
MS	0.084295	0.004674	18.03667	0.0000***
IR	60.37884	11.37890	5.306212	0.0000***
IPI	-28.10266	5.362852	-5.240246	0.0000***
R-squared	0.924967	Mean dependent var		1515.094
Adjusted R-squared	0.921705	S.D. dependent var		314.6340
F-statistic	283.5334	Durbin-Watson stat		0.698711
Prob(F-statistic)	0.000000	AIC		11.84174

Note: ***p-value < 0.01

Table 4.4.1: Summary of OLS model using HAC standard errors and covariance.

4.5. Lagged Dependent Variables Model

The lag order required to address the autocorrelation is identified using the Akaike Information Criterion (AIC) and the Hannan-Quinn Criterion (HQC). The AIC and HQC vary as we included different lag orders in the model, and the models with the lowest AIC and HQC suggest the appropriate lag order to incorporate. As shown in Table 4.5.1, the model should follow a first-order autocorrelation. Therefore, GF(-1) is incorporated to estimate the model more precisely. As displayed in Table 4.5.2, the results indicate that the autocorrelation problem is resolved in both the Durbin-Watson test, with a Durbin-Watson statistic of nearly 2, and the Breusch-Godfrey test, with a p-value of 0.7065. Thus, by adding the GF(-1), we addressed the autocorrelation problem. The model produced a slightly different result from the previous models. The USD has become insignificant, with a p-value of 0.0869, while GF(-1)

is highly significant in the model. Additionally, the significance of the remaining variables weakens but remains significant at the 0.01 level of significance. The model's explanatory power has improved from 0.924967 to 0.967034, and the AIC is lowered from 11.84174 to 11.04053. Therefore, we can conclude that the lagged dependent variable HAC model is the best model.

Lag order	Akaike Info Criterion	Hannan-Quinn Criterion
1	11.04053*	11.10656*
2	11.06392	11.13979
3	11.08777	11.17357
4	11.09388	11.18973
5	11.11114	11.21714

Table 4.5.1: Different lag-order models' AIC and HQC.

Variables	Coefficient	Std. Error	T-statistic	p-value
C	1253.144	377.1737	3.322458	0.0012***
USD	-2.713414	1.571384	-1.726767	0.0869*
CPI	-1.726025	14.95310	-0.115429	0.9083
MS	0.018555	0.005954	3.116190	0.0023***
IR	19.29180	5.979998	3.226054	0.0016***
IPI	-9.150430	2.346014	-3.900416	0.0002***
GF(-1)	0.773424	0.063623	12.15635	0.0000***
R-squared	0.967034	Mean dependent var		1517.022
Adjusted R-squared	0.965284	S.D. dependent var		315.2351

F-statistic	552.4673	Durbin-Watson stat	2.018622
Prob(F-statistic)	0.000000	AIC	11.04053
Breusch-Godfrey Test			
F-statistic	0.348500	Prob. F	0.7065
Obs*R-squared	0.748811	Prob. Chi-Square	0.6877

Note: *p-value < 0.1, ***p-value < 0.01.

Table 4.5.2: Lagged dependent variable model summary.

4.6. Results Interpretation

The model suggests that H1 is not supported, but USD is significant at the 10% level. The coefficient argues that when the US dollar index increases by one, the gold futures price decreases by approximately \$2.7134 on average, holding other variables constant. H2 is not supported in our analysis. Our findings conclude that H3 is supported, as MS is significant at 1% level. Holding other variables constant, a one billion increase in the U.S. money supply will cause the gold futures price to increase by approximately \$0.0186, on average. Additionally, IR is discovered to be significant at 1% level, supporting H4. When the federal funds effective rate increased by 100 basis points, the gold futures price increased by approximately \$19.2918, on average, while holding other variables constant. Moreover, H5 is supported as IPI is significant at 1% level. By holding other variables constant, an increase in the industrial production index will lead to a decrease in the gold futures price of approximately \$9.1504, on average. Lastly, the gold futures price is found to depend on the previous month's price with a 99% confidence level. A one-dollar increase in the past month's gold futures price will lead to an approximately \$0.7734 increase in the gold futures price on average by holding other variables constant.

5. Conclusion

In our study, we have determined that the first lagged dependent variable HAC model is the best model free from issues. The result supported H3, H4, and H5. MS are proven to positively influence the gold futures price. This aligns with the liquidity preference theory, which proposes that as the money supply increases, the currency depreciates and investors shift towards safer and more liquid gold investments. This will drive the gold and gold futures price up.

Besides, IR is positively affecting the gold futures price. However, this is in the opposite direction of where we argue, according to the liquidity preference theory, that higher interest rates should lead to a lower gold futures price because they will make non-yielding gold investments less attractive than interest-bearing assets. The negative relationship can be justified using the Efficient Market Hypothesis (EMH), which states that when economic crises occur, gold investments, which act as a risk hedge, will become popular and rise in price. This is precisely true when the interest rate is the critical instrument for the Federal Reserve to curb high inflation. According to Wang et al. (2019), the price of gold is influenced by inflation, where higher inflation leads to increased demand for gold as a hedge against inflation. Additionally, the federal funds rate is positively correlated with U.S. inflation. As inflation rises, the Federal Reserve is likely to increase the federal funds rate, and the gold price is expected to rise due to economic uncertainty.

On the other hand, IPI is negatively influencing GF. As suggested by Rana and Connor (2023), this negative relationship is due to the capital flow to productive sectors when the economy is booming, which causes the demand for gold to drop. As high IPI indicates favourable economic production, the optimistic information should be reflected in the market according to the EMH. This would lead to a decline in gold prices because other investments, such as stocks, tend to outperform gold investments during economic booms.

Moreover, GF are found to be related to the last month's gold futures price. This finding confirms the argument by Ntim et al. (2015) that the gold market does not fully follow the Random Walk Hypothesis and can be predicted using past and current information. However, the predictive power of past information stops at the first lagged order as GF only displays a significant relationship with the previous month's GF.

USD is discovered to be insignificant, rejecting H1. However, it is significant at the 0.1 level

of significance, suggesting a possible negative relationship with GF. This conclusion is justified by the fact that gold futures are typically quoted in US dollars per ounce. Batten et al. (2010) and O'Connor and Lucey (2012) explained that strengthening the US dollar makes the gold price more expensive to foreign investors, thereby lowering their demand for gold and the gold futures price. On the other hand, H2 is rejected as CPI is statistically insignificant. This may be due to a few reasons. Firstly, gold price is affected primarily by high inflation, but moderate or declining inflation does not influence the price materially (*CPI And Gold - Simple Explanation*, n.d.). Additionally, gold's effectiveness as an inflation hedge is evident in the long run, where short-term price level fluctuations do not significantly impact the gold price (FocusEconomics, 2025). Furthermore, the gold futures market is only sensitive to unexpected changes in forecasted inflation (Christie-David et al., 2000). Thus, the CPI growth for the past period is irrelevant to predicting the gold futures price.

(6,474 words)

6. Reference

- Abdullah, A., & Bakar, M. J. A. (2015). The application of gold price, interest rates and inflation expectations in capital markets. *International Journal of Economics and Finance*, 7(2). <https://doi.org/10.5539/ijef.v7n2p293>
- Ambarita, S., & Djamaluddin, S. (2020, September 20). Analysis of Macroeconomic Effects on Gold Price. *IOSR Journal of Business and Management (IOSR-JBM)* e-ISSN: 2278-487X, p-ISSN: 2319-7668. Volume 22, Issue 9. Ser. V (September 2020), PP 47-57. DOI: 10.9790/487X-2209054757
- Artigas, J. C. (2010). *Linking global money supply to gold and gold to future inflation*. World Gold Council. Retrieved from https://www.spdrgoldshares.com/media/GLD/file/money_supply_paper_jan10.pdf
- Athina, K. (2024). *The Conditional Relationships between Gold and Stock Market Prices: Evidence from USA*. DIVA. <http://hj.diva-portal.org/smash/record.jsf?pid=diva2:1866688>
- Baklaci, H. F., Süer, Ö., & Yelkenci, T. (2016). PRICE LINKAGES AMONG EMERGING GOLD FUTURES MARKETS. *The Singapore Economic Review*, 63(05), 1345–1365. <https://doi.org/10.1142/s021759081650020x>
- Batten, J. A., Ciner, C., & Lucey, B. M. (2010). The macroeconomic determinants of volatility in precious metals markets. *Resources Policy*, 35(2), 65–71. <https://doi.org/10.1016/j.resourpol.2009.12.002>
- Batten, J. A., Ciner, C., & Lucey, B. M. (2014). On the economic determinants of the gold–inflation relation. *Resources Policy*, 41, 101–108. <https://doi.org/10.1016/j.resourpol.2014.03.007>
- Baur, D. G., & Lucey, B. M. (2010). *Is Gold a Hedge or a Safe Haven? An Analysis of Stocks, Bonds and Gold*. *Financial Review*, 45(2), 217–229.
- Ben A. H., Boubaker S., Ftiti Z., Louhichi W., Tissaoui (2024). Forecasting commodity prices: empirical evidence using deep learning tools. *Annals of Operations Research*, 2024, 339(1): 349-67.
- Bibow, J. (2013). *Keynes on monetary policy, finance and uncertainty*.

<https://doi.org/10.4324/9780203170533>

BullionVault. (n.d.). *Gold futures explained*. BullionVault.
<https://www.bullionvault.com/gold-guide/gold-futures>

Chang, X. (2024). Gold price inflation and its implications for financial markets: Strategies for mitigation. *Advances in Economics Management and Political Sciences*, 119(1), 189–194. <https://doi.org/10.54254/2754-1169/119/20242530>

Chiang, T. C. (2022). The effects of economic uncertainty, geopolitical risk and pandemic upheaval on gold prices. *Resources Policy*, 76, 102546.
<https://doi.org/10.1016/j.resourpol.2021.102546>

Choong, P. S., Kwo, P. Y., Piong, C. K., & Wong, W. X. (2012, May). Determinants of Gold Price: Using Simple And Multiple Linear Regression. University Tunku Abdul Rahman. <http://eprints.utar.edu.my/1097/1/BF-2012-0905944.pdf>

Christie-David, R., Chaudhry, M., & Koch, T. W. (2000). Do macroeconomics news releases affect gold and silver prices? *Journal of Economics and Business*, 52, 405–421.

CME Group. (n.d.). *Gold Overview - CME Group*. @CMEGroup.
<https://www.cmegroup.com/markets/metals/precious/gold.html>

CPI and Gold - Simple Explanation. (n.d.).
<https://www.goldpriceforecast.com/explanations/cpi-gold/>

Dierinck, B., Frömmel, M., Schrauwen, B., & Hermans, M. (2012). Determinants of the gold price. Unievrsiteit Gent. https://libstore.ugent.be/fulltxt/RUG01/001/893/290/RUG01-001893290_2012_0001_AC.pdf

Dinh, T., Goutte, S., Nguyen, D. K., & Walther, T. (2022). Economic drivers of volatility and correlation in precious metal markets. *Journal of Commodity Markets*, 28, 100242.
<https://doi.org/10.1016/j.jcomm.2021.100242>

Erb, C. B., & Harvey, C. R. (2013). The golden dilemma. *Financial Analysts Journal*, 69, 10–42.

Erdoğan, N. A. (2017). The most significant factors influencing the price of gold: An empirical analysis of the US market. *Economics World*, 5(5). <https://doi.org/10.17265/2328->

- Ergül, Ö., & Karakaş, T. (2024). Analysis of the relationship of gold prices with inflation and bitcoin in the post-tapering period. *Borsa Istanbul Review*, 24(4), 797–805. <https://doi.org/10.1016/j.bir.2024.04.009>
- Fama, E. F. (1991). Efficient Capital Markets: II. *The Journal of Finance*, 46(5), 1575–1617. <https://doi.org/10.1111/j.1540-6261.1991.tb04636.x>
- Fan, W., Fang, S., & Lu, T. (2014, February 11). Macro-factors on gold pricing during the financial crisis. *China Finance Review International*, Vol. 4 No. 1, pp. 58-75. <https://doi.org/10.1108/CFRI-09-2012-0097>
- Fang, L., Yu, H., & Xiao, W. (2018). Forecasting gold futures market volatility using macroeconomic variables in the United States. *Economic Modelling*, 72, 249–259. <https://doi.org/10.1016/j.econmod.2018.02.003>
- FocusEconomics. (2025, May 6). *Gold, inflation and interest rates: how do they interact?* FocusEconomics. <https://www.focus-economics.com/blog/gold-inflation-interest-rates-prices-impact/>
- FRED. (n.d.a). *Nominal Broad U.S. Dollar Index*. FRED, Federal Reserve Bank of St. Louis. <https://fred.stlouisfed.org/series/TWEXBGSMTN>
- FRED. (n.d.b). *Consumer Price Index: All Items: Total for United States*. Stlouisfed.org. <https://fred.stlouisfed.org/series/CPALTT01USM657N>
- FRED. (n.d.c). *M2 Money Stock*. Stlouisfed.org. <https://fred.stlouisfed.org/series/M2SL>
- FRED. (n.d.d). *Effective Federal Funds Rate*. Stlouisfed.org. <https://fred.stlouisfed.org/series/FEDFUNDS>
- FRED. (n.d.e). *Industrial Production Index*. Stlouisfed.org. <https://fred.stlouisfed.org/series/INDPRO>
- Ghosh, D., Levin, E. J., Macmillan, P., & Wright, R. E. (2004). *Gold as an Inflation Hedge?*. *Studies in Economics and Finance*, 22(1), 1–25.
- Gold Price Forecast (n.d.). *Industrial production and gold - explained*. Gold Price Forecast.

- <https://www.goldpriceforecast.com/explanations/industrial-production-gold/>
- Gurufocus.com. (2025). *US Dollar Index (DXY) Charts, Data*. Gurufocus.com.
https://www.gurufocus.com/economic_indicators/5747/us-dollar-index-dxy
- Hashim, S. L., Ramlan, H., Razali, N. H., & Nordin, N. Z. (2017). Macroeconomic variables affecting the volatility of gold price. *Journal of Global Business and Social Entrepreneurship (GBSE)*, 3(5), 97-106. <http://gbse.my/v3no5may17/Paper-83-.pdf>
- He, T. (n.d.). *Gold Futures Price Forecast Based on Artificial Intelligence*. Ewadirect.
<https://www.ewadirect.com/proceedings/aemps/article/view/21823/pdf>
- Hergt, B. (2013). Gold prices during and after the Great Recession. *U.S. Bureau of Labor Statistics*, 2(4). <https://www.bls.gov/opub/btn/volume-2/pdf/gold-prices-during-and-after-the-great-recession.pdf>
- Honor. (2024, March 12). *The Consumer Price Index (CPI) and gold price*. Tavex Bullion.
<https://tavexbullion.co.uk/introduction-to-consumer-price-index-and-gold-price/>
- Ibrahim, S. N., Kamaruddin, N. I., & Hasan, R. (2014). The determinants of gold prices in Malaysia. *Journal of Advanced Management Science*, 2(1).
<https://www.joams.com/uploadfile/2013/1225/20131225043038701.pdf>
- Industrial production and gold - explained.* (n.d.-b).
<https://www.goldpriceforecast.com/explanations/industrial-production-gold/>
- Investing.com (n.d.a). *Gold Futures Streaming Chart*. Investing.com.
<https://www.investing.com/commodities/gold-streaming-chart>
- Investing.com (n.d.b). *XAU/USD Streaming Chart*. Investing.com.
<https://www.investing.com/currencies/xau-usd-chart>
- Investing.com. (2025). *Gold Futures Historical Prices*. Investing.com.
<https://www.investing.com/commodities/gold-historical-data>
- Jabeur, S. B., Mefteh-Wali, S., & Viviani, J.-L. (2024). Forecasting gold price with the XGBoost algorithm and SHAP interaction values. *Annals of Operations Research*, 334(1), 679–699.
- Jia, Y., Dong, Z., & An, H. (2023). Study of the Modal Evolution of the Causal Relationship

- between Crude Oil, Gold, and Dollar Price Series. *International Journal of Energy Research*, 2023, 1–15. <https://doi.org/10.1155/2023/7947434>
- Jin, M., Li, Y., Wang, J., & Yang, Y. C. (2018). Price discovery in the Chinese gold market. *Journal of Futures Markets*, 38(10), 1262–1281. <https://doi.org/10.1002/fut.21938>
- Kamran, A., Israr, S., & Rizvi, S. M. A. (2013). Determinants of gold prices in Pakistan. In *Lecture notes in electrical engineering* (pp. 1219–1227). https://doi.org/10.1007/978-3-642-40081-0_103
- Kang, S. L. (2022, March 21). *Gold price's fate tied to Russia-Ukraine conflict, say analysts*. TheEdgeMalaysia. <https://theedgemalaysia.com/article/gold-prices-fate-tied-russiaukraine-conflict-say-analysts>
- Kasteler, A. (2017). Quantitative Easing: Money Supply and the Commodity Prices of Oil, Gold, and Wheat. *All Graduate Plan B and other Reports*. 1037. <https://digitalcommons.usu.edu/gradreports/1037>
- Kim, A. (2023). Creating Volatility from Gold Prices. *SSRN Electronic Journal*. <https://doi.org/10.2139/ssrn.4620035>
- Lampinen, A. (2007). Gold investments and short and long run price determinants of the price of gold. Lappeenranta University of Technology. <https://lutpub.lut.fi/bitstream/handle/10024/30680/TMP.objres.688.pdf?sequence=1>
- Levin, E.J., Montagnoli, A. and Wright, R.E. (2006) Short-run and long-run determinants of the price of gold. World Gold Council. <https://strathprints.strath.ac.uk/7215/>
- Liu, S., Tang, T., McKenzie, A. M., & Liu, Y. (2015). Low-Frequency volatility in China's gold futures market and its macroeconomic determinants. *Mathematical Problems in Engineering*, 2015, 1–8. <https://doi.org/10.1155/2015/646239>
- Lucey, B. M., Larkin, C. J., & O'Connor, F. A. (2012). London or New York: Where and when does the gold price originate? *SSRN Electronic Journal*. <https://doi.org/10.2139/ssrn.2161905>
- Mamcarz, K. (2015). Determinants of gold price in the short term. *Zeszyty Naukowe Uniwersytetu Szczecińskiego Finanse Rynki Finansowe Ubezpieczenia*, 2015(74/1), 127–137. <https://doi.org/10.18276/frfu.2015.74/1-11>

- MCB Group. (2025). *Financial market analysis: The 2025 gold rush*. MCB Group.
<https://mcbgroup.com/think/article/financial-market-analysis--the-2025-gold-rush>
- Mo, D., Gupta, R., Li, B., & Singh, T. (2017). The macroeconomic determinants of commodity futures volatility: Evidence from Chinese and Indian markets. *Economic Modelling*, 70, 543–560. <https://doi.org/10.1016/j.econmod.2017.08.032>
- Ntim, C. G., English, J., Nwachukwu, J., & Wang, Y. (2015). On the efficiency of the global gold markets. *International Review of Financial Analysis*, 41, 218–236. <https://doi.org/10.1016/j.irfa.2015.03.013>
- Nylund, Drake A. 2017. A Look into the Determinants of the Price of Gold”. University of Nevada, Reno
- O'Connor, F., & Lucey, B. (2012). Gold's negative relationship to the US dollar. *The Alchemist*, 66, 16
- Ogiriki, T., & Andabai, P. (2014). Liquidity Preference Theory: A comparison of William Baumol's and James Tobin's propositions. *African Research Review*, 8(4), 143. <https://doi.org/10.4314/afrrrev.v8i4.12>
- Oka, T. (2021). Interpretation of Chapter 17 of The General Theory and reconciliation between the endogenous money supply and the liquidity preference theory. *The Japanese Political Economy*, 47(2–3), 261–280. <https://doi.org/10.1080/2329194x.2021.1984253>
- Radev, L., Golitsis, P., & Mitreva, M. (2023). Economic and financial determinants of gold ETF price volatility on the U.S. Futures Market (COMEX). *Journal of Economics*, 8(2), 12–26. <https://doi.org/10.46763/joe2382012r>
- Raju, G. A., & Shirodkar, S. (2020). THE LEAD LAG RELATIONSHIP BETWEEN SPOT AND FUTURES MARKETS IN THE ENERGY SECTOR: EMPIRICAL EVIDENCE FROM INDIAN MARKETS. *International Journal of Energy Economics and Policy*, 10(5), 409–414. <https://doi.org/10.32479/ijeep.9783>
- Rana, H. M. U. (2022). The Macroeconomic Determinants and Market Efficiency of Precious Metals: An Empirical Evidence of International Markets. <https://www.open-access.bcu.ac.uk/id/eprint/14444>

- Rana, H. M. U., & O'Connor, F. (2023). Domestic macroeconomic determinants of precious metals prices in developed and emerging economies: An international analysis of the long and short run. *International Review of Financial Analysis*, 89, 102813. <https://doi.org/10.1016/j.irfa.2023.102813>
- Risse, M. (2019). Combining wavelet decomposition with machine learning to forecast gold returns. *International Journal of Forecasting*, 35(2), 601–615. <https://doi.org/10.1016/j.ijforecast.2018.11.008>
- Seemuang, Ananya & Romprsert, Suppanunta. 2013. “Gold Value Movement and Macroeconomics”. *Journal of Business and Economics*, USA, Volume 4, No. 8, pp. 752-760, ISSN 2155-7950
- Shubik, M., & Yao, S. (1990). Gold, liquidity and secured loans in a multistage economy. *Journal of Economics*, 52(1), 1–23. <https://doi.org/10.1007/bf01227499>
- Silva, Emmanuel S. (2014). Forecasting the price of gold. *Atlantic Economic Journal*, 14, 43–52.
- Šimáková, J. (2011). Analysis of the relationship between oil and gold prices. *Journal of finance*, 51(1), 651-662.
- Sindhu, S. D. (2013). A study on impact of select factors on the price of Gold. *IOSR Journal of Business and Management*, 8(4), 84–93. <https://doi.org/10.9790/487x-0848493>
- Singh, A., & Kaur, N. (2020). Macroeconomic Determinants of Gold Prices : A bounds Testing approach. *Indian Journal of Finance*, 14(2), 21. <https://doi.org/10.17010/ijf/2020/v14i2/150554>
- Sipkova, Hana and Sipko, Juraj. 2014. “What Are the Real Drivers of Gold Price?” *International Journal of Business and Social Science* Vol.5, No. 8(1).
- Stetler, M. (2025, April 16). *Gold vs. Money Supply - Vaulted*. Vaulted. <https://vaulted.com/nuggets/gold-vs-money-supply/>
- StonexBullion. (2025, February 25). *Gold Prices vs. Interest Rates - What You Need to Know*. StonexBullion. <https://stonexbullion.com/en/blog/gold-price-and-interest-rate-relationship/>

- Sun, Y. (2018). The Impact of Macroeconomic Factors on the Price of Gold. *Applied Economics Theses*, 34. https://digitalcommons.buffalostate.edu/cgi/viewcontent.cgi?article=1036&context=economics_theses
- Swissbullion. (2025, March 11). *How the US Dollar Affects Gold Prices*. Swissbullion. <https://www.swissbullion.eu/en/posts/how-does-the-strength-of-the-us-dollar-affect-gold-prices>
- Synek, R. (2024). Cointegration analysis of US M2 and gold price over the last half century. *European Financial and Accounting Journal*. Retrieved from <https://efaj.vse.cz/pdfs/efa/2024/01/01.pdf>
- Tantatape, K., & Komain, J. (2007). Cointegration and Causality between Stock Index and Macroeconomic Variables in Emerging Market. *Academy of Accounting and Financial Studies Journal*, 11(3).
- Team, C. (2024, May 22). *Efficient Markets Hypothesis*. Corporate Finance Institute. <https://corporatefinanceinstitute.com/resources/career-map/sell-side/capital-markets/efficient-markets-hypothesis/>
- Țițan, A. G. (2015). The Efficient Market Hypothesis: Review of specialized literature and empirical research. *Procedia Economics and Finance*, 32, 442–449. [https://doi.org/10.1016/s2212-5671\(15\)01416-1](https://doi.org/10.1016/s2212-5671(15)01416-1)
- Toraman, C., Basarir, C., & Bayramoglu, M. F. (2011). Determination of factors affecting the price of gold: a study of MGARCH model. *Business and Economics Research Journal*, 2(4), 1–37. <https://www.acarindex.com/dosyalar/makale/acarindex-1423873750.pdf>
- Valadkhani, A., Nguyen, J., & Chiah, M. (2022). When is gold an effective hedge against inflation? *Resources Policy*, 79, 103009. <https://doi.org/10.1016/j.resourpol.2022.103009>
- Wang, F., Chang, J., Zuo, W., & Zhou, W. (2024). Research on Efficiency and Multifractality of Gold Market under Major Events. *Fractal and Fractional*, 8(8), 488. <https://doi.org/10.3390/fractalfract8080488>
- Wang, H., Sheng, H., & Zhang, H. (2019). Influence factors of international gold futures price

- volatility. *Transactions of Nonferrous Metals Society of China*, 29(11), 2447–2454.
[https://doi.org/10.1016/s1003-6326\(19\)65151-4](https://doi.org/10.1016/s1003-6326(19)65151-4)
- Wang, Y. S., & Chueh, Y. L. (2012). Dynamic transmission effects between the interest rate, the US dollar, and gold and crude oil prices. *Economic Modelling*, 30, 792–798.
<https://doi.org/10.1016/j.econmod.2012.09.052>
- WSJ Staff. (2025, April 21). *Gold Price Surges to Another Record High*. The Wall Street Journal. <https://www.wsj.com/livecoverage/stock-market-trump-tariffs-trade-war-04-21-25/card/gold-price-surges-to-another-record-high-ZtfX9WTrtBGlyqu5BJPx>
- Zhang, P., & Ci, B. (2020). Deep belief network for gold price forecasting. *Resources Policy*, 69, 101806. <https://doi.org/10.1016/j.resourpol.2020.101806>
- Zhang, Z., & Qin, Y. (2022). Study on the nonlinear interactions among the international oil price, the RMB exchange rate and China's gold price. *Resources Policy*, 77, 102683. <https://doi.org/10.1016/j.resourpol.2022.102683>

7. Appendix

Descriptive Statistics

	GF	USD	CPI	MS	IR	IPI
Mean	1515.094	113.4488	0.236810	16146.12	1.397603	100.7898
Median	1327.300	114.4106	0.215078	14526.20	0.650000	101.6718
Maximum	2227.100	127.4828	1.373608	21749.60	5.330000	104.1038
Minimum	1060.200	93.60390	-0.668694	11219.10	0.050000	84.68120
Std. Dev.	314.6340	7.046144	0.363256	3725.859	1.656724	3.018109
Skewness	0.390485	-1.058257	0.339990	0.326980	1.294159	-2.586150
Kurtosis	1.648291	4.356005	3.628129	1.457508	3.509228	12.93454
Jarque-Bera	12.28671	31.85516	4.320290	14.15169	35.08346	632.4655
Probability	0.002148	0.000000	0.115308	0.000845	0.000000	0.000000
Sum	183326.4	13727.31	28.65397	1953681.	169.1100	12195.57
Sum Sq. Dev.	11879349	5957.778	15.83460	1.67E+09	329.3682	1093.078
Observations	121	121	121	121	121	121

Linear Model

Dependent Variable: GF
Method: Least Squares
Date: 06/18/25 Time: 20:24
Sample: 2014M01 2024M01
Included observations: 121

Variable	Coefficient	Std. Error	t-Statistic	Prob.
C	4486.428	469.0723	9.564470	0.0000
USD	-13.98623	1.904190	-7.344974	0.0000
CPI	10.25255	25.61427	0.400267	0.6897
MS	0.084295	0.003389	24.87482	0.0000
IR	60.37884	7.581121	7.964369	0.0000
IPI	-28.10266	3.495941	-8.038654	0.0000
R-squared	0.924967	Mean dependent var	1515.094	
Adjusted R-squared	0.921705	S.D. dependent var	314.6340	
S.E. of regression	88.03846	Akaike info criterion	11.84174	
Sum squared resid	891338.5	Schwarz criterion	11.98037	
Log likelihood	-710.4252	Hannan-Quinn criter.	11.89804	
F-statistic	283.5334	Durbin-Watson stat	0.698711	
Prob(F-statistic)	0.000000			

Correlation Matrix

Correlation						
	GF	USD	CPI	MS	IR	IPI
GF	1.000000	0.579904	0.373673	0.931052	0.473781	-0.082224
USD	0.579904	1.000000	0.105459	0.683209	0.545068	-0.135748
CPI	0.373673	0.105459	1.000000	0.410782	0.010172	0.068306
MS	0.931052	0.683209	0.410782	1.000000	0.447966	0.001665
IR	0.473781	0.545068	0.010172	0.447966	1.000000	0.447772
IPI	-0.082224	-0.135748	0.068306	0.001665	0.447772	1.000000

Auxiliary Regressions

Dependent Variable: USD
Method: Least Squares
Date: 06/18/25 Time: 21:23
Sample: 2014M01 2024M01
Included observations: 121

Variable	Coefficient	Std. Error	t-Statistic	Prob.
C	176.0833	15.99476	11.00881	0.0000
CPI	-1.632836	1.239708	-1.317114	0.1904
MS	0.000959	0.000139	6.885370	0.0000
IR	2.007906	0.319198	6.290472	0.0000
IPI	-0.799000	0.153471	-5.206185	0.0000
R-squared	0.641210	Mean dependent var	113.4488	
Adjusted R-squared	0.628838	S.D. dependent var	7.046144	
S.E. of regression	4.292725	Akaike info criterion	5.792165	
Sum squared resid	2137.589	Schwarz criterion	5.907693	
Log likelihood	-345.4260	Hannan-Quinn criter.	5.839085	
F-statistic	51.82732	Durbin-Watson stat	0.161054	
Prob(F-statistic)	0.000000			

Dependent Variable: CPI
Method: Least Squares
Date: 06/18/25 Time: 21:32
Sample: 2014M01 2024M01
Included observations: 121

Variable	Coefficient	Std. Error	t-Statistic	Prob.
C	-1.575012	1.694013	-0.929752	0.3544
USD	-0.009024	0.006851	-1.317114	0.1904
MS	6.27E-05	1.08E-05	5.799321	0.0000
IR	-0.055420	0.026994	-2.053011	0.0423
IPI	0.018854	0.012551	1.502238	0.1358
R-squared	0.253940	Mean dependent var	0.236810	
Adjusted R-squared	0.228214	S.D. dependent var	0.363256	
S.E. of regression	0.319125	Akaike info criterion	0.593979	
Sum squared resid	11.81356	Schwarz criterion	0.709508	
Log likelihood	-30.93575	Hannan-Quinn criter.	0.640900	
F-statistic	9.870877	Durbin-Watson stat	1.099939	
Prob(F-statistic)	0.000001			

Dependent Variable: MS
Method: Least Squares
Date: 06/18/25 Time: 21:36
Sample: 2014M01 2024M01
Included observations: 121

Variable	Coefficient	Std. Error	t-Statistic	Prob.
C	-18871.26	12731.95	-1.482197	0.1410
USD	302.6627	43.95736	6.885370	0.0000
CPI	3583.481	617.9138	5.799321	0.0000
IR	302.6013	205.8036	1.470340	0.1442
IPI	-5.862328	95.78256	-0.061205	0.9513
R-squared	0.594839	Mean dependent var	16146.12	
Adjusted R-squared	0.580868	S.D. dependent var	3725.859	
S.E. of regression	2412.136	Akaike info criterion	18.45486	
Sum squared resid	6.75E+08	Schwarz criterion	18.57039	
Log likelihood	-1111.519	Hannan-Quinn criter.	18.50178	
F-statistic	42.57653	Durbin-Watson stat	0.243632	
Prob(F-statistic)	0.000000			

Dependent Variable: IR
Method: Least Squares
Date: 06/18/25 Time: 21:36
Sample: 2014M01 2024M01
Included observations: 121

Variable	Coefficient	Std. Error	t-Statistic	Prob.
C	-43.13182	4.118928	-10.47161	0.0000
USD	0.126677	0.020138	6.290472	0.0000
CPI	-0.632646	0.308155	-2.053011	0.0423
MS	6.05E-05	4.11E-05	1.470340	0.1442
IPI	0.291018	0.033213	8.762291	0.0000
R-squared	0.590554	Mean dependent var	1.397603	
Adjusted R-squared	0.576435	S.D. dependent var	1.656724	
S.E. of regression	1.078227	Akaike info criterion	3.028956	
Sum squared resid	134.8584	Schwarz criterion	3.144485	
Log likelihood	-178.2519	Hannan-Quinn criter.	3.075877	
F-statistic	41.82746	Durbin-Watson stat	0.194131	
Prob(F-statistic)	0.000000			

Dependent Variable: IPI
Method: Least Squares
Date: 06/18/25 Time: 21:37
Sample: 2014M01 2024M01
Included observations: 121

Variable	Coefficient	Std. Error	t-Statistic	Prob.
C	125.6194	4.377673	28.69548	0.0000
USD	-0.237050	0.045532	-5.206185	0.0000
CPI	1.012147	0.673760	1.502238	0.1358
MS	-5.51E-06	9.00E-05	-0.061205	0.9513
IR	1.368543	0.156186	8.762291	0.0000
R-squared	0.419816	Mean dependent var	100.7898	
Adjusted R-squared	0.399810	S.D. dependent var	3.018109	
S.E. of regression	2.338187	Akaike info criterion	4.577073	
Sum squared resid	634.1858	Schwarz criterion	4.692602	
Log likelihood	-271.9129	Hannan-Quinn criter.	4.623994	
F-statistic	20.98418	Durbin-Watson stat	0.388472	
Prob(F-statistic)	0.000000			

Variance Inflation Factor

Variance Inflation Factors
Date: 06/19/25 Time: 19:29
Sample: 2014M01 2024M01
Included observations: 121

Variable	Coefficient Variance	Uncentered VIF	Centered VIF
C	220028.8	3434.948	NA
USD	3.625941	731.3403	2.787149
CPI	656.0909	1.914760	1.340375
MS	1.15E-05	49.20511	2.468156
IR	57.47339	4.194896	2.442326
IPI	12.22161	1939.941	1.723592

Heteroscedasticity Test Results

Heteroskedasticity Test: Breusch-Pagan-Godfrey
Null hypothesis: Homoskedasticity

F-statistic	3.410511	Prob. F(5,115)	0.0066
Obs*R-squared	15.62529	Prob. Chi-Square(5)	0.0080
Scaled explained SS	18.48418	Prob. Chi-Square(5)	0.0024

Test Equation:
Dependent Variable: RESID^2
Method: Least Squares
Date: 06/19/25 Time: 19:16
Sample: 2014M01 2024M01
Included observations: 121

Variable	Coefficient	Std. Error	t-Statistic	Prob.
C	46735.45	60804.05	0.768624	0.4437
USD	285.4949	246.8329	1.156632	0.2498
CPI	-1370.344	3320.280	-0.412719	0.6806
MS	0.515881	0.439273	1.174398	0.2427
IR	-575.3337	982.7117	-0.585455	0.5594
IPI	-783.4019	453.1655	-1.728732	0.0865
R-squared	0.129135	Mean dependent var	7366.434	
Adjusted R-squared	0.091271	S.D. dependent var	11971.48	
S.E. of regression	11412.09	Akaike info criterion	21.57105	
Sum squared resid	1.50E+10	Schwarz criterion	21.70968	
Log likelihood	-1299.048	Hannan-Quinn criter.	21.62735	
F-statistic	3.410511	Durbin-Watson stat	1.163137	
Prob(F-statistic)	0.006551			

White's Test

Heteroskedasticity Test: White
Null hypothesis: Homoskedasticity

F-statistic	3.627257	Prob. F(20,100)	0.0000
Obs*R-squared	50.87342	Prob. Chi-Square(20)	0.0002
Scaled explained SS	60.18150	Prob. Chi-Square(20)	0.0000

Test Equation:
Dependent Variable: RESID^2
Method: Least Squares
Date: 06/19/25 Time: 19:31
Sample: 2014M01 2024M01
Included observations: 121

Variable	Coefficient	Std. Error	t-Statistic	Prob.
C	-4875573.	4650675.	-1.048358	0.2970
USD^2	-46.10280	57.99689	-0.794919	0.4285
USD*CPI	-969.1475	816.6922	-1.186674	0.2382
USD*MS	0.597558	0.196955	3.033984	0.0031
USD*IR	-179.5421	544.0365	-0.330019	0.7421
USD*IPI	-81.35793	244.9770	-0.332104	0.7405
USD	10981.21	33136.88	0.331389	0.7410
CPI^2	-1281.458	9161.996	-0.139867	0.8890
CPI*MS	2.391355	1.727102	1.384605	0.1693
CPI*IR	1727.771	3322.393	0.520038	0.6042
CPI*IPI	620.8694	1474.058	0.421197	0.6745
CPI	8805.424	202235.8	0.043540	0.9654
MS^2	-0.001456	0.000434	-3.354070	0.0011
MS*IR	0.450749	0.894797	0.503745	0.6155
MS*IPI	-0.520467	0.387707	-1.342422	0.1825
MS	31.97676	42.58261	0.750935	0.4545
IR^2	1243.872	1309.323	0.950011	0.3444
IR*IPI	401.0943	1176.877	0.340813	0.7340
IR	-39257.83	145994.6	-0.268899	0.7886
IPI^2	-302.4372	202.7975	-1.491327	0.1390
IPI	78096.86	62637.03	1.246816	0.2154
R-squared	0.420442	Mean dependent var	7366.434	
Adjusted R-squared	0.304530	S.D. dependent var	11971.48	
S.E. of regression	9983.600	Akaike info criterion	21.41176	
Sum squared resid	9.97E+09	Schwarz criterion	21.89698	
Log likelihood	-1274.412	Hannan-Quinn criter.	21.60883	
F-statistic	3.627257	Durbin-Watson stat	1.814259	
Prob(F-statistic)	0.000010			

BG Test

Breusch-Godfrey Serial Correlation LM Test:
Null hypothesis: No serial correlation at up to 8 lags

F-statistic	11.68843	Prob. F(8,107)	0.0000
Obs*R-squared	56.42882	Prob. Chi-Square(8)	0.0000

Test Equation:
Dependent Variable: RESID
Method: Least Squares
Date: 06/19/25 Time: 20:09
Sample: 2014M01 2024M01
Included observations: 121
Presample missing value lagged residuals set to zero.

Variable	Coefficient	Std. Error	t-Statistic	Prob.
C	-536.1793	399.5993	-1.341792	0.1825
USD	0.924494	1.469343	0.629189	0.5306
CPI	17.85462	20.21983	0.883025	0.3792
MS	-0.001666	0.002669	-0.624157	0.5339
IR	0.339272	5.797635	0.058519	0.9534
IPI	4.516249	3.109691	1.452314	0.1493
RESID(-1)	0.726983	0.097079	7.488547	0.0000
RESID(-2)	-0.023166	0.120267	-0.192625	0.8476
RESID(-3)	-0.143017	0.118162	-1.210350	0.2288
RESID(-4)	0.193608	0.120280	1.609638	0.1104
RESID(-5)	0.092883	0.119610	0.776545	0.4391
RESID(-6)	-0.185300	0.118811	-1.559621	0.1218
RESID(-7)	0.228880	0.120647	1.897116	0.0605
RESID(-8)	-0.081043	0.106779	-0.758980	0.4495
R-squared	0.466354	Mean dependent var	-9.42E-13	
Adjusted R-squared	0.401518	S.D. dependent var	86.18481	
S.E. of regression	66.67394	Akaike info criterion	11.34595	
Sum squared resid	475659.3	Schwarz criterion	11.66943	
Log likelihood	-672.4299	Hannan-Quinn criter.	11.47733	
F-statistic	7.192879	Durbin-Watson stat	1.836843	
Prob(F-statistic)	0.000000			

HAC standard errors and covariance model with AR1 scheme

Dependent Variable: GF
Method: Least Squares
Date: 06/20/25 Time: 14:42
Sample: 2014M01 2024M01
Included observations: 121

Variable	Coefficient	Std. Error	t-Statistic	Prob.
C	4486.428	469.0723	9.564470	0.0000
USD	-13.98623	1.904190	-7.344974	0.0000
CPI	10.25255	25.61427	0.400267	0.6897
MS	0.084295	0.003389	24.87482	0.0000
IR	60.37884	7.581121	7.964369	0.0000
IPI	-28.10266	3.495941	-8.038654	0.0000
R-squared	0.924967	Mean dependent var	1515.094	
Adjusted R-squared	0.921705	S.D. dependent var	314.6340	
S.E. of regression	88.03846	Akaike info criterion	11.84174	
Sum squared resid	891338.5	Schwarz criterion	11.98037	
Log likelihood	-710.4252	Hannan-Quinn criter.	11.89804	
F-statistic	283.5334	Durbin-Watson stat	0.698711	
Prob(F-statistic)	0.000000			

Lagged Dependent Variable HAC model

First lag

Dependent Variable: GF
Method: Least Squares
Date: 06/20/25 Time: 14:48
Sample (adjusted): 2014M02 2024M01
Included observations: 120 after adjustments
HAC standard errors & covariance (Bartlett kernel, Newey-West fixed bandwidth = 5.0000)

Variable	Coefficient	Std. Error	t-Statistic	Prob.
C	1253.144	377.1737	3.322458	0.0012
USD	-2.713414	1.571384	-1.726767	0.0869
CPI	-1.726025	14.95310	-0.115429	0.9083
MS	0.018555	0.005954	3.116190	0.0023
IR	19.29180	5.979998	3.226054	0.0016
IPI	-9.150430	2.346014	-3.900416	0.0002
GF(-1)	0.773424	0.063623	12.15635	0.0000
R-squared	0.967034	Mean dependent var	1517.022	
Adjusted R-squared	0.965284	S.D. dependent var	315.2351	
S.E. of regression	58.73542	Akaike info criterion	11.04053	
Sum squared resid	389833.0	Schwarz criterion	11.20313	
Log likelihood	-655.4315	Hannan-Quinn criter.	11.10656	
F-statistic	552.4673	Durbin-Watson stat	2.018622	
Prob(F-statistic)	0.000000	Wald F-statistic	802.2475	
Prob(Wald F-statistic)	0.000000			

Second Lag

Dependent Variable: GF
 Method: Least Squares
 Date: 06/20/25 Time: 16:16
 Sample (adjusted): 2014M03 2024M01
 Included observations: 119 after adjustments

Variable	Coefficient	Std. Error	t-Statistic	Prob.
C	1220.968	422.2102	2.891848	0.0046
USD	-2.491427	1.661254	-1.499727	0.1365
CPI	-1.488355	17.58708	-0.084628	0.9327
MS	0.018022	0.006192	2.910322	0.0044
IR	19.00132	6.197373	3.066028	0.0027
IPI	-9.072060	2.846554	-3.187032	0.0019
GF(-1)	0.750511	0.101564	7.389550	0.0000
GF(-2)	0.028195	0.096944	0.290842	0.7717
R-squared	0.966987	Mean dependent var	1518.880	
Adjusted R-squared	0.964906	S.D. dependent var	315.9074	
S.E. of regression	59.18059	Akaike info criterion	11.06392	
Sum squared resid	388759.9	Schwarz criterion	11.25076	
Log likelihood	-650.3035	Hannan-Quinn criter.	11.13979	
F-statistic	464.4786	Durbin-Watson stat	1.969790	
Prob(F-statistic)	0.000000			

Third lag

Dependent Variable: GF
 Method: Least Squares
 Date: 06/20/25 Time: 16:18
 Sample (adjusted): 2014M04 2024M01
 Included observations: 118 after adjustments

Variable	Coefficient	Std. Error	t-Statistic	Prob.
C	1252.214	429.5969	2.914858	0.0043
USD	-2.723003	1.728393	-1.575454	0.1180
CPI	-1.705683	18.13603	-0.094049	0.9252
MS	0.018414	0.006523	2.822954	0.0057
IR	19.28378	6.275420	3.072906	0.0027
IPI	-9.133993	2.876707	-3.175156	0.0019
GF(-1)	0.751353	0.102372	7.339438	0.0000
GF(-2)	0.044045	0.124625	0.353425	0.7245
GF(-3)	-0.020177	0.099211	-0.203372	0.8392
R-squared	0.966865	Mean dependent var	1521.192	
Adjusted R-squared	0.964433	S.D. dependent var	316.2414	
S.E. of regression	59.64100	Akaike info criterion	11.08777	
Sum squared resid	387718.3	Schwarz criterion	11.29909	
Log likelihood	-645.1784	Hannan-Quinn criter.	11.17357	
F-statistic	397.5658	Durbin-Watson stat	1.941832	
Prob(F-statistic)	0.000000			

Lagged Dependent Variable Model's Breusch-Godfrey test

First lag

Breusch-Godfrey Serial Correlation LM Test:
Null hypothesis: No serial correlation at up to 2 lags

F-statistic	0.348500	Prob. F(2,111)	0.7065
Obs*R-squared	0.748811	Prob. Chi-Square(2)	0.6877

Test Equation:

Dependent Variable: RESID

Method: Least Squares

Date: 06/20/25 Time: 15:53

Sample: 2014M02 2024M01

Included observations: 120

Presample missing value lagged residuals set to zero.

Variable	Coefficient	Std. Error	t-Statistic	Prob.
C	-126.9400	444.6605	-0.285476	0.7758
USD	0.497127	1.716521	0.289613	0.7727
CPI	1.855334	17.52379	0.105875	0.9159
MS	-0.003935	0.007586	-0.518745	0.6050
IR	-2.110338	6.657748	-0.316975	0.7519
IPI	0.664352	2.949391	0.225251	0.8222
GF(-1)	0.046075	0.084964	0.542290	0.5887
RESID(-1)	-0.090133	0.122722	-0.734450	0.4642
RESID(-2)	-0.068732	0.112443	-0.611259	0.5423
R-squared	0.006240	Mean dependent var	-5.07E-14	
Adjusted R-squared	-0.065382	S.D. dependent var	57.23554	
S.E. of regression	59.07701	Akaike info criterion	11.06760	
Sum squared resid	387400.4	Schwarz criterion	11.27666	
Log likelihood	-655.0560	Hannan-Quinn criter.	11.15250	
F-statistic	0.087125	Durbin-Watson stat	1.949907	
Prob(F-statistic)	0.999492			